Chapter 1

Indefinite Integrals

Definition: The notation, $\int f(x)\,dx$, It is traditionally used for an antiderivative of f and is called an indefinite integral.

Table of integrals:

Expression	Formula
$\int \sin x\,dx$	$-\cos x + c$
$\int \cos x\,dx$	$\sin x + c$
$\int \sec^2(x)\,dx$	$\tan x + c$
$\int \csc^2 x\,dx$	$-\cot x + c$
$\int tan(x)\sec(x)dx$	$\sec x + c$
$\int cosec(x)\cot(x)\,dx$	$-\operatorname{cosec} x + c$
$\int \mathrm{x}^{\mathrm{n}} dx$	$\frac{x^{n+1}}{n+1} + c$
$\int \frac{1}{\mathrm{x}} dx$	$\ln x + c$
$\int \sqrt{x} dx$	$\frac{2}{3}x^{\frac{3}{2}} + c$
$\int \frac{dx}{\sqrt{x}}$	$2\sqrt{x} + c$
$\int a^x dx$	$\frac{a^x}{\ln a} + c$
$\int \frac{1}{\mathrm{a}^2+x^2} dx$	$\frac{1}{a}\tan^{-1}\frac{x}{a} + c$
$\int \frac{1}{\sqrt{a^2-x^2}} dx$	$\sin^{-1}\frac{x}{a} + c$
$\int e^{mx}\,dx$	$\frac{e^{mx}}{m} + c$
$\int \sin mx\,dx$	$-\frac{\cos mx}{m} + c$
$\int \cos mx\,dx$	$\frac{\sin mx}{m} + c$
$\int \frac{f'(x)}{f(x)} dx$	$\ln\lvert f(x)\rvert + c$

Example 1 **(a)** Evaluate $\int \frac{\cos x}{\sin^2 x}\,dx$

Solution:

$$\int \frac{\cos x}{\sin^2 x}\,dx$$
$$= \int \frac{\cos x}{\sin x}\,\frac{1}{\sin x} dx$$
$$= \int \csc x \cot x\,dx$$
$$= -\csc x + c$$

(b) Find $\int \frac{t^2-2t^4}{t^4}\,dt$

$$\int \frac{t^2-2t^4}{t^4}\,dt$$
$$= \int \frac{t^2}{t^4} - \frac{2t^4}{t^4}\,dt$$
$$= \int (t^{-2} - 2)\,dt$$
$$= \frac{t^{-1}}{-1} - 2t + c$$

Exercise Set 1.1

Evaluate the indefinite integrals.

1. $\int (5 + \frac{2}{3}x^2 + \frac{3}{4}x^3)\, dx$
2. $\int \left(u^6 - 2u^5 - u^3 + \frac{2}{7}\right) du$
3. $\int (u + 4)(2u + 1) du$
4. $\int \sqrt{t}\,(t^2 + 3t + 2) dt$
5. $\int \frac{1+\sqrt{x}+x}{x}\, dx$
6. $\int \left(x^2 + 1 + \frac{1}{x^2+1}\right) dx$
7. $\int (\sin x + \sinh x)\, dx$
8. $\int \left(\frac{1+r}{r}\right)^2 dr$
9. $\int (2 + \tan^2\theta) d\theta$
10. $\int \sec t\,(\sec t + \tan t)\, dt$
11. $\int 2^t(1 + 5^t)\, dt$
12. $\int \frac{\sin 2x}{\sin x} dx$
13. $\int \left(\sqrt{x} - 3x^{\frac{7}{5}} + \frac{1}{9}\right) dx$
14. $\int \left(\frac{10}{y^{\frac{3}{5}}} - \sqrt[3]{y} + \frac{4}{\sqrt{y}}\right) dy$
15. $\int (\csc^2 t - \sec t \tan t) dt$
16. $\int \frac{\sec\theta}{\cos\theta} d\theta$
17. $\int \frac{dy}{\csc y}$
18. $\int \left(\phi + \frac{2}{\sin^2\phi}\right) d\phi$
19. $\int \frac{\sec x + \cos x}{2\cos x} dx$
20. $\int (1 + \sin^2\theta \csc\theta) d\theta$

[1-12] [Ref. Calculus-J. Stewart 8th ed.: Page-409], **[13-20]** [Ref. Calculus-H. Anton 10th ed.: Page-279]

The Substitution Rule: If $u = g(x)$ is a differentiable function whose range is an interval *I* and f is continuous on *I*, then $\int f(g(x))g'(x)\,dx = \int f(u)\,du$.

Example 2: Find $\int x^3 \cos(x^4+2)\,dx$.

Solution: $\int x^3 \cos(x^4+2)\,dx$

$= \int \cos u \frac{1}{4} du = \frac{1}{4}\int \cos u\,du$

$= \frac{1}{4}\sin u + c = \frac{1}{4}\sin(x^4+2) + c$

Let, $u = x^4 + 2$

$du = 4x^3 dx$

$x^3 dx = \frac{1}{4}\,du$

Example 3: Find $\int \frac{x}{\sqrt{1-4x^2}}\,dx$.

Solution: $\int \frac{x}{\sqrt{1-4x^2}}\,dx$

$= -\frac{1}{8}\int \frac{1}{\sqrt{u}}\,du = -\frac{1}{8}\int u^{-\frac{1}{2}}\,du$

$= -\frac{1}{8}\,2\sqrt{u} + C = -\frac{1}{4}\sqrt{1-4x^2} + C$

Let, $u = 1 - 4x^2$

$du = -8x\,dx$

$x\,dx = -\frac{1}{8}\,du$

- More examples 5,6,8 (Calculus-J. Stewart 8th ed.: Page-415) & 3-5 (Calculus-H. Anton 10th ed.: Page-283)

Exercise Set 1.2

Evaluate the indefinite integrals.

1. $\int x\sqrt{1-x^2}\,dx$
2. $\int x^2 e^{x^3}\,dx$
3. $\int \sin t\,(1+\cos t)\,dt$
4. $\int \cos\frac{\pi t}{2}\,dt$
5. $\int \sec^2 2\theta\,d\theta$
6. $\int y(4-y^2)^{\frac{2}{3}}\,dy$
7. $\int \cos^3\theta \sin\theta\,d\theta$
8. $\int \frac{e^u}{(1-e^u)^2}\,du$
9. $\int \frac{\sin\sqrt{x}}{\sqrt{x}}\,dx$
10. $\int \frac{z^2}{z^3+1}\,dz$
11. $\int \frac{(\ln x)^2}{x}\,dx$
12. $\int \sin x \sin(\cos x)\,dx$
13. $\int x\sqrt{x+2}\,dx$
14. $\int \frac{dx}{\sqrt{1-x^2}\,\sin^{-1}x}$
15. $\int x^2\sqrt{1+x}\,dx$
16. $\int \frac{6}{(1-2x)^3}\,dx$
17. $\int \frac{x^2+1}{\sqrt{x^3+3x}}\,dx$
18. $\int \frac{\sin\left(\frac{5}{x}\right)}{x^2}\,dx$
19. $\int \frac{\sec^2(\sqrt{x})}{\sqrt{x}}\,dx$
20. $\int \frac{y}{\sqrt{2y+1}}\,dy$

[1-14] [Ref. Calculus-J. Stewart 8th ed.: Page-418-419], **[15-20]** [Ref. Calculus- H. Anton 10th ed.: page-286]

Chapter 2

Definite Integrals

The Fundamental Theorem of Calculus: If f is continuous on $[a, b]$, then $\int_a^b f(x)\ dx = F(b) - F(a)$.

Where F is any antiderivative of f, that is, a function such that $F' = f$.

Example 1: Evaluate the integral $\int_1^3 e^x\ dx$.

Solution: The function $f(x) = e^x$ is continuous everywhere, the antiderivative of the function is $F(x) = e^x$.

So, $\int_1^3 e^x\ dx = F(3) - F(1) = e^3 - e$.

Example 2: Find the area under the parabola $y = x^2$ from 0 to 1.

Solution: An antiderivative of the function $f(x) = x^2$ is $F(x) = \frac{1}{3}x^3$. From the **Fundamental Theorem,**

$$\int_0^1 x^2\ dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3}.$$

Exercise Set 2.1

1. $\int_0^1 (1 - 8v^3 + 16v^7)\, dv$

2. $\int_0^4 (4 - t)\sqrt{t}\, dt$

3. $\int_1^4 \frac{2+x^2}{\sqrt{x}} dx$

4. $\int_{-1}^2 (3u - 2)(u + 1)\ du$

5. $\int_1^4 \frac{v^3+3v^6}{v^4}\ dv$

6. $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{8}{1+x^2}\ dx$

7. $\int_1^2 \left(x^{-7} - x\sqrt{x}\right)\ dx$

8. $\int_0^1 (5x - 5^x)\, dx$

9. $\int_{-2}^0 \left(\frac{1}{2}t^4 + \frac{1}{3}t^3 - t\right)\ dt$

10. $\int_0^{\pi} (5e^x + \sin x)\ dx$

11. $\int_1^2 \left(\frac{1}{x^2} - \frac{4}{x^3}\right)\ dx$

12. $\int_1^4 \frac{4+6u}{\sqrt{u}}\ du$

13. $\int_1^2 \left(\frac{x}{2} - \frac{2}{x}\right) dx$

14. $\int_0^{\pi} f(x)\, dx$, where $f(x) = \begin{cases} \sin x, \text{ if } 0 \le x \le \frac{\pi}{2} \\ \cos x, \text{ if } \frac{\pi}{2} \le x \le \pi \end{cases}$

15. If oil leaks from a tank at a rate of $r(t)$ gallons per minute at time t, what does $\int_0^{120} r(t)\, dt$ represent?

16. The marginal cost of manufacturing x yards of a certain fabric is, $C'(x) = 3 - 0.01x + 0.000006x^2$ in dollars per yard). Find the increase in cost if the production level is raised from 2000 yards to 4000 yards.

17. A spring follows Hooks Law ($F = kx$) where $k = 200\ N/m$. The spring is stretched from its natural length to $0.4m$. Determine the work done in this process.

18. The electric current (I) in a circuit varies with time (t) given by the equation $I(t) = 3t^2 + 2t + 1$. The circuit is controlled by a switch. For a test, the switch was turned on and then turned off after 5 seconds. Find the total electric charge that passed through the circuit between these 5 seconds.

[1-15] [Ref. Calculus- J. Stewart 8th ed.: Page, 409,419]

Substitution for Definite integrals: If $u = g(x)$ is a differentiable function whose range is an interval *I* and f is continuous on *I*, then $\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$

Example 3: Evaluate $\int_0^4 \sqrt{2x+1}\,dx$.

Solution: $\int_0^4 \sqrt{2x+1}\,dx$
$= \frac{1}{2}\int_1^9 \sqrt{u}\,du$
$= \frac{1}{2}\cdot\left[\frac{2}{3}\,u^{\frac{3}{2}}\right]_1^9$
$= \frac{1}{3}\,(9^{\frac{3}{2}} - 1)$
$= \frac{26}{3}$

Let, $u = 2x+1$
$du = 2dx$
$dx = \frac{1}{2}\,du$

When,
$x = 0,\ u = 1$
$x = 4,\ u = 9$

Example 4: Evaluate $\int_1^3 \frac{\cos\left(\frac{\pi}{x}\right)}{x^2}dx$.

Solution: $\int_1^3 \frac{\cos\left(\frac{\pi}{x}\right)}{x^2}dx$
$= -\frac{1}{\pi}\int_{\frac{\pi}{3}}^{\pi}\cos u\,du$
$= -\left[\frac{1}{\pi}\sin u\right]_{\frac{\pi}{3}}^{\pi}$
$= -\frac{1}{\pi}\,(\sin\frac{\pi}{3} - 0)$
$= -\frac{\sqrt{3}}{2\pi}$

Let, $u = \frac{\pi}{x}$
$du = -\pi\cdot\frac{1}{x^2}\,dx$
$\frac{1}{x^2}dx = -\frac{1}{\pi}du$

When,
$x = 1,\ u = \pi$
$x = 3,\ u = \frac{\pi}{3}$

Exercise Set 2.2

1. $\int_0^1 (3t-1)^{50}dt$
2. $\int_0^1 \sqrt{1+7x}\,dx$
3. $\int_0^{\frac{\pi}{6}} \frac{\sin t}{\cos^2 t}dt$
4. $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2}\,dx$
5. $\int_0^2 (x-1)\,e^{(x-1)^2}dx$
6. $\int_0^1 \frac{e^z+1}{e^z+z}\,dz$
7. $\int_{-\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\sin x}{\sqrt{2+\cos x}}\,dx$
8. $\int_{-3}^0 x\sqrt{1-x}\,dx$
9. $\int_{1-\pi}^{1+\pi} \sec^2\left(\frac{x}{4}-\frac{1}{4}\right)dx$
10. $\int_1^2 \frac{dx}{x^2-6x+9}$
11. A particle moves with a velocity of $v(t) = \sin\pi t\ \ m/s$ along an s-axis. Find the distance traveled by the particle over the time interval $0 \le t \le 1$.
12. Wastage and pollutant materials are dumped into a river at a rate $R(t) = 100(1-e^{-0.1t})$kg/day. What will be the total waste dumped into the river in a December month?
13. The **rate of absorption** of an antibiotic drug in the bloodstream is $r(t) = 165e^{-0.3t}$mg/hour. A patient took a tablet, and it took 8 hours to completely absorbed in blood. Find the weight of the tablet.
14. The cost per meter to build a beam is $C(x) = 6x^2\sqrt{1+x^3}$ dollars where $0 \le x \le 3$. Find the total cost to build it.

[1-6] [Ref. Calculus- J. Stewart 8th ed.: Page-419], **[7-11]** [Ref. Calculus- H. Anton 10th ed.: Page-34]

Integrals of Symmetric Functions: Suppose f is continuous on $[-a, a]$.

(a) If f is even $[f(-x) = f(x)]$, then $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$.

(b) If f is odd $[f(-x) = -f(x)]$, then $\int_{-a}^{a} f(x)\,d = 0$

Example 5: Evaluate $\int_{-2}^{2}(x^6+1)\ dx$.

Solution: Since $f(x) = x^6 + 1$ satisfies $f(-x) = f(x)$, it is even and so,

$$\int_{-2}^{2}(x^6+1)\ dx\ = 2\int_0^2 (x^6+1)\ dx == 2\left[\frac{x^7}{7}+x\right]_0^2 = 2\left(\frac{128}{7}+2\right) = \frac{284}{7}$$

Exercise Set 2.3

Evaluate the integrals. [1-4] [Ref. Calculus- J. Stewart 8th ed.: Page-419,422]

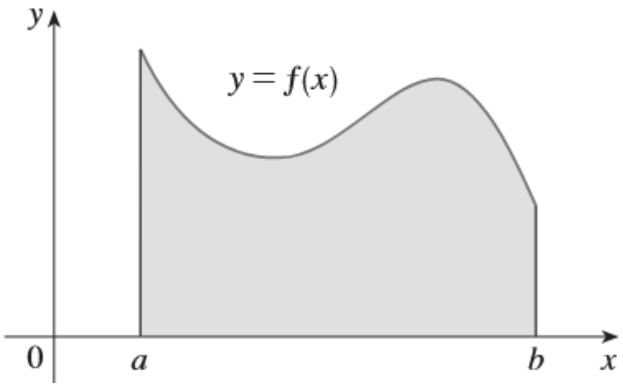
1. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (x^3 + x^4\tan x)\,dx$
2. $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} x^4\sin x\,dx$
3. $\int_{-1}^{1} \frac{\sin x}{1+x^2}\,dx$
4. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{t^4\tan t}{2+\cos t}\,dt$

Riemann Sum

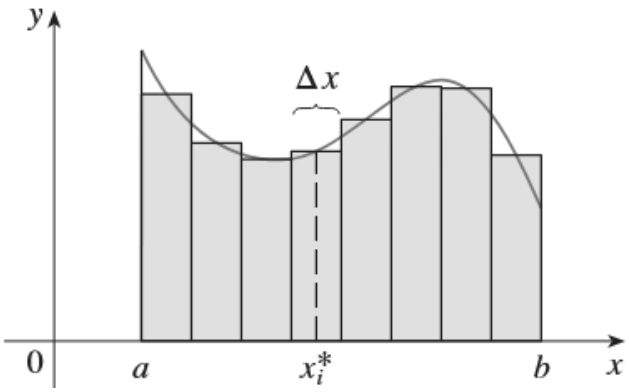
If f is a function defined for $a < x < b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = \frac{b-a}{n}$. We let $x_0\ (= a),\ x_1(= a + \Delta x), x_2\ (= a + 2 \cdot \Delta x), \cdots, x_n\ (= b)$ be the endpoints of these subintervals and we let $x_1^*,\ x_2^*, \cdots, x_n^*$ be any sample points in these subintervals, so x_i^* lies in the i-th subinterval $[x_{i-1}, x_i]$. The definite integral of f from a to b is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x \qquad \text{..... (1)}$$

The sum that occurs in (1) $\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$ is called a Riemann sum after the German mathematician **Bernhard Riemann (1826 –1866)**. So, the definite integral of an integrable function can be approximated to within any desired degree of accuracy by a Riemann sum.

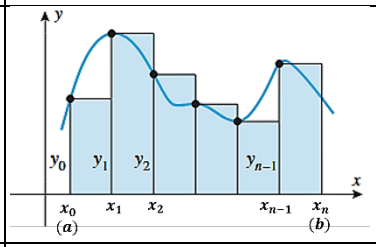
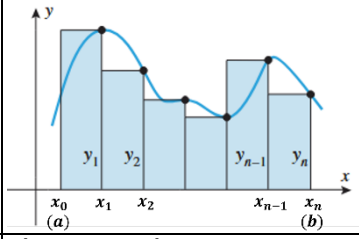
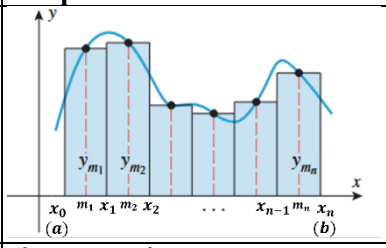


The integral $\int_a^b f(x)\,dx$ is the area under the curve $y = f(x)$ from a to b.



The Riemann sum $\sum f(x_i^*)\Delta x$ is the sum of areas of rectangles.

There are 3 different approaches for finding the Riemann Sum approximations. Those are,

Rules	**Left End Point Rule**	**Right End Point Rule**	**Midpoint Rule**
Figure			
Formula	$\int_a^b f(x)\,dx = \frac{b-a}{n}[y_0 + y_1 + \cdots + y_{n-1}]$ here, $y_i = f(x_i)$	$\int_a^b f(x)\,dx = \frac{b-a}{n}[y_1 + y_2 + \cdots + y_n]$ here, $y_i = f(x_i)$	$\int_a^b f(x)\,dx = \frac{b-a}{n}[y_{m_1} + y_{m_2} + \cdots + y_{m_n}]$ here, $m_i = \frac{x_{i-1}+x_i}{2}$ and $y_{m_i} = f(m_i)$

Among the methods, The Midpoint Rule is mathematically the most accurate one among the three mentioned rules.

Example 6: Given, $f(x) = x^3 - 6x$, with $a = 0$, $b = 3$, and $n = 6$.

(a) Evaluate the Riemann sum for the given function by taking the sample points to be left endpoints.

(b) Evaluate the Riemann sum for the given function by taking the sample points to be right endpoints.

Solution: With $n = 6$ the interval width is $\Delta x = \frac{b-a}{n} = \frac{3-0}{6} = \frac{1}{2} = 0.5$.

(a) The left endpoints are,
$x_i = 0, 0.5, 1, 1.5, 2, 2.5$
So, the Riemann Sum is,

$$\sum_{i=1}^{6} f(x_i)\Delta x$$
$$= \Delta x[f(0) + f(0.5) + f(1) + f(1.5) + f(2) + f(2.5)]$$
$$= \frac{1}{2}[0 + (-2.875) + (-5) + (-5.625) + (-4) + 0.625]$$
$$= -8.4375$$

(b) The right endpoints are,
$x_i = 0.5, 1, 1.5, 2, 2.5, 3$
So, the Riemann Sum is,

$$\sum_{i=0}^{5} f(x_i)\Delta x$$
$$= \Delta x[f(0.5) + f(1) + f(1.5) + f(2) + f(2.5) + f(3)]$$
$$= \frac{1}{2}[(-2.875) + (-5) + (-5.625) + (-4) + 0.625 + 9]$$
$$= -3.9375$$

Example 7: Use the Midpoint Rule with $n = 5$ to approximate $\int_1^2 \frac{1}{x} dx$.

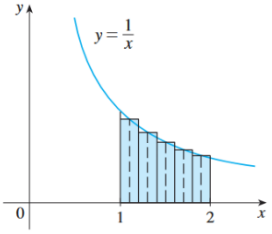
Solution: We have here is $a = 1$, $b = 2$ and $n = 5$. So, $\Delta x = \frac{b-a}{n} = \frac{2-1}{5} = \frac{1}{5} = 0.2$.

So, the endpoints are,
$x_i = 1, 1.2, 1.4, 1.6, 1.8, 2$

Based on the endpoints, the midpoints are,
$m_1 = \frac{x_0+x_1}{2} = \frac{1+1.2}{2} = \frac{2.2}{2} = 1.1$
$m_2 = 1.3,\ m_3 = 1.5, m_4 = 1.7, m_5 = 1.9$

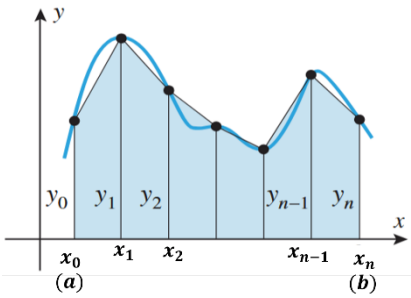
So, using the Midpoint rule of Riemann sum, we get,

$$\int_1^2 \frac{1}{x} dx = \sum_{i=0}^{5} f(m_i)\Delta x$$
$$= f(1.1)\Delta x + f(1.3)\Delta x + f(1.5)\Delta x + f(1.7)\Delta x + f(1.9)\Delta x$$
$$= \Delta x[f(1.1) + f(1.3) + f(1.5) + f(1.7) + f(1.9)]$$
$$= \frac{1}{5}\left[\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9}\right] \approx 0.69108$$



Trapezoidal Rule

Let us denote, left endpoint and right endpoint approximations with n subintervals by L_n, R_n respectively. If we take the average of L_n and R_n, then we obtain another important approximation denoted by $T_n = \frac{1}{2}(L_n + R_n)$; which is known as the Trapezoidal Rule or Trapezoidal approximation.



If we need to approximate, $\int_a^b f(x)\,dx$ then the Trapezoidal Rule formula:
Here, $\Delta x = \frac{b-a}{n}$.
So, $\int_a^b f(x)\,dx \approx T_n = \frac{\Delta x}{2}[y_0 + 2\cdot y_1 + 2\cdot y_2 + \cdots + 2\cdot y_{n-1} + y_n]$;
where, $y_i = f(x_i)$.

Example 8: Use the Trapezoidal Rule with $n = 5$ to approximate $\int_1^2 \frac{1}{x}dx$.

Solution: We have here is $a = 1$, $b = 2$ and $n = 5$. So, $\Delta x = \frac{b-a}{n} = \frac{2-1}{5} = \frac{1}{5} = 0.2$. So, the Trapezoidal rule gives,

$\int_1^2 \frac{1}{x}dx \approx T_5 = \frac{0.2}{2}[f(1) + 2\cdot f(1.2) + 2\cdot f(1.4) + 2\cdot f(1.6) + 2\cdot f(1.8) + f(2)]$

$= 0.1\cdot\left[\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2}\right] \approx 0.695635.$

Exercise 2.4

1. Evaluate the Riemann sum (using right endpoint rules) for $f(x) = x - 1, -6 \leq x \leq 4$, with five subintervals.
2. If $f(x) = \cos x, 0 \leq x \leq \frac{3\pi}{4}$, find the Riemann sum with $n = 6$, taking the sample points to be left endpoints.
3. If $f(x) = x^2 - 4, 0 \leq x \leq 3$, find the Riemann sum with $n = 6$, taking the sample points to be midpoints.
4. Use Midpoint rule to approximate $\int_0^{\pi} x \sin^2 x \, dx$ with $n = 4$, Round the answer to four decimal places.
5. Estimate the following integrals using Left Endpoint, Right Endpoint, Midpoint and Trapezoidal Rule.
   i. $\int_1^2 \sqrt{x^3 - 1}\, dx, n = 6$
   ii. $\int_0^2 \frac{e^x}{1+x^2} dx, n = 5.$
   iii. $\int_0^{\frac{\pi}{2}} \sqrt{1 + \cos x}\, dx, n = 4.$
   iv. $\int_0^4 \sqrt{y} \cos y \, dy, n = 5.$
   v. $\int_2^3 \frac{1}{\ln t} dt, n = 6.$
   vi. $\int_1^3 \frac{\sin t}{t} dt, n = 4.$
   vii. $\int_0^1 \sqrt{x + x^3}\, dt, n = 6.$
   viii. $\int_0^1 \cos(x^2)\, dx, \ n = 4.$
   ix. $\int_0^2 \frac{x}{1+x^2} dx, n = 6.$
   x. $\int_0^{\pi} x \cos x \, dx, \ n = 4.$

[1-5] [Ref: Calculus - James Stewart – 8th Ed.: Page: 388]

Chapter 3

Techniques of Integration

Integration by Parts: Every differentiation rule has a corresponding integration rule. For instance, the Substitution Rule for integration corresponds to the Chain Rule for differentiation. The rule that corresponds to the Product Rule for differentiation is called the rule for **integration by parts**.

$$\int uv\,dx = u\int v dx - \int \left\{\frac{du}{dx}\cdot\int v dx\right\}dx$$

The sequence of assuming two functions u and v (where u and v are functions of x) in an integral is significantly important. There is a technique that helps to choose u and v properly in this method.

The technique is known as **LIATE**. This acronym comes from the first letter of the following types of functions.

L: **Logarithm** I: **Inverse trigonometric** A: **Algebraic** T: **Trigonometric** E: **Exponential**

Example 1: Find $\int x\sin x\,dx$.

Solution: Let, $u = x \quad v = \sin x$

$$\int x\sin x\,dx = x\cdot\int \sin x\,dx - \int\left\{\frac{dx}{dx}\int \sin x\,dx\right\}dx$$
$$= -x\cos x + \int \cos x\,dx$$
$$= -x\cos x + \sin x + c$$

Therefore, $\int x\sin x\,dx = -x\cos x + \sin x + c$

Example 2: Find $\int \ln x\,dx$.

Solution: Let, $u = \ln x,\ v = 1$

$$\int \ln x\,dx = \ln x\cdot\int dx - \int\left\{\frac{d}{dx}(\ln x)\int dx\right\}dx$$
$$= x\cdot\ln x - \int\frac{1}{x}\cdot x\,dx$$
$$= x\ln x - \int dx$$
$$= x\ln x - x + c$$

Therefore, $\int \ln x\,dx = x\ln x - x + c$

Example 3: Find $\int x^2e^{3x}dx$.

Solution: Here $u = x^2,\ v = e^{3x}$

$$\int x^2e^{3x}dx = x^2\int e^{3x}dx - \int\left(\frac{d}{dx}x^2\int e^{3x}dx\right)dx$$
$$= x^2\frac{1}{3}e^{3x} - \int\left(2x\frac{1}{3}e^{3x}\right)dx$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{3}\int x\cdot e^{3x}\,dx$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{3}\left[x\int e^{3x}dx - \int\left(\frac{d}{dx}x\int e^{3x}dx\right)dx\right]$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{3}\left[x\frac{1}{3}e^{3x} - \int 1\cdot\frac{1}{3}e^{3x}\,dx\right]$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}x\,e^{3x} + \frac{2}{9}\int e^{3x}\,dx$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}x\,e^{3x} + \frac{2}{9}\cdot\frac{1}{3}e^{3x}$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}x\,e^{3x} + \frac{2}{27}e^{3x} + c$$

Another technique: **Tabular method**

+	x^2	e^{3x}
−	$2x$	$\frac{1}{3}e^{3x}$
+	2	$\frac{1}{9}e^{3x}$
−	0	$\frac{1}{27}e^{3x}$

$$\int x^2e^{3x}dx = x^2\frac{1}{3}e^{3x} - 2x\cdot\frac{1}{9}e^{3x} + 2\cdot\frac{1}{27}e^{3x} + c$$
$$= \frac{1}{3}x^2e^{3x} - \frac{2}{9}x\,e^{3x} + \frac{2}{27}e^{3x} + c$$

Exercise Set 3.1

Evaluate the integrals.

1. $\int xe^{2x}dx$
2. $\int \sqrt{x}\ln x\,dx$
3. $\int x\cos 5x\,dx$
4. $\int te^{-3t}dx$
5. $\int (x-1)\sin \pi x\,dx$
6. $\int (x^2+2x)\cos x\,dx$
7. $\int t^2\sin \beta t\,dt$
8. $\int \cos^{-1} x\,dx$
9. $\int \ln\sqrt{x}\,dx$
10. $\int t^4\ln t\,dt$
11. $\int z^3e^z dz$
12. $\int_0^{1/2} x\cos \pi x\,dx$
13. $\int x^2e^{-2x}\,dx$
14. $\int x\ln x\,dx$
15. $\int (\ln x)^2\,dx$
16. $\int \sin^{-1} x\,dx$
17. $\int x\tan^{-1} x\,dx$
18. $\int xe^{(x-1)}dx$
19. $\int x^5\ln x\,dx$
20. $\int_0^2 \tan^{-1} 2x\,dx$
21. $\int (2x^2+1)\cos 2x\,dx$

[1-12] [Ref: Calculus – J. Stewart – 8th Ed.: Page: 476], **[13-21]** [Ref: Calculus – H. Anton – 10th Ed.: Page: 498]

Trigonometric Integrals:

$\int \sin^m x \cos^n x\, dx$	Procedure	Relevant Identities
n odd	Split off a factor of $\cos x$. Apply the relevant identity. Make the substitution $z = \sin x$	$\cos^2 x = 1 - \sin^2 x$
m odd	Split off a factor of $\sin x$. Apply the relevant identity. Make the substitution $z = \cos x$	$\sin^2 x = 1 - \cos^2 x$
m even	Use the relevant identities to reduce the powers on $\sin x$ and $\cos x$	$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$
n even		$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

Integrand	Relevant Trigonometric Identity
$\int \sin(mx)\cos(nx)\, dx$	$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$
$\int \cos(mx)\sin(nx)\, dx$	$\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$
$\int \sin(mx)\sin(nx)\, dx$	$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$
$\int \cos(mx)\cos(nx)\, dx$	$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$

Example 4: Find $\int \sin^4 x\, dx$.

Solution:
$$\begin{aligned}
&\int \sin^4 x\, dx \\
&= \int (\sin^2 x)^2\, dx \\
&= \int \left(\tfrac{1-\cos 2x}{2}\right)^2 dx \\
&= \tfrac{1}{4}\int (1 - 2\cos 2x + \cos^2 2x)\, dx \\
&= \tfrac{1}{4}\int \left[1 - 2\cos 2x + \tfrac{1}{2}(1 + \cos 4x)\right] dx \\
&= \tfrac{1}{4}\int \left(\tfrac{3}{2} - 2\cos 2x + \tfrac{1}{2}\cos 4x\right) dx \\
&= \tfrac{1}{4}\left[\tfrac{3}{2}x - \sin 2x + \tfrac{1}{8}\sin 4x\right] + c \\
&= \tfrac{3}{8}x - \tfrac{1}{4}\sin 2x + \tfrac{1}{32}\sin 4x + c
\end{aligned}$$

Example 5: Evaluate $\int \sin^4 x \cos^5 x\, dx$.

Solution:
$$\begin{aligned}
&\int \sin^4 x \cos^5 x\, dx \\
&= \int \sin^4 x \cos^4 x \cos x\, dx \\
&= \int \sin^4 x\, (\cos^2 x)^2 \cos x\, dx \\
&= \int \sin^4 x\, (1 - \sin^2 x)^2 \cos x\, dx \\
&= \int z^4 (1 - z^2)^2 dz \\
&= \int z^4 (1 - 2z^2 + z^4) dz \\
&= \int (z^4 - 2z^6 + z^8) dz \\
&= \tfrac{z^5}{5} - \tfrac{2z^7}{7} + \tfrac{z^9}{9} + c \\
&= \tfrac{1}{5}\sin^5 x - \tfrac{2}{7}\sin^7 x + \tfrac{1}{9}\sin^9 x + c
\end{aligned}$$

Example 6: Evaluate $\int \sin 7x \cos 3x\, dx$.

Solution:
$$\begin{aligned}
&\int \sin 7x \cos 3x\, dx \\
&= \int \tfrac{1}{2}[\sin(7+3)x + \sin(7-3)x] dx \\
&= \tfrac{1}{2}\int [\sin 10x + \sin 4x] dx \\
&= \tfrac{1}{2}\left[-\tfrac{1}{10}\cos 10x - \tfrac{1}{4}\cos 4x\right] + c \\
&= -\tfrac{1}{20}\cos 10x - \tfrac{1}{8}\cos 4x + c
\end{aligned}$$

Example 7: Evaluate $\int \sin 4x \cos 5x\, dx$.

Solution:
$$\begin{aligned}
&\int \sin 4x \cos 5x\, dx \\
&= \int \tfrac{1}{2}[\sin(4+5)x + \sin(4-5)x] dx \\
&= \tfrac{1}{2}\int [\sin 9x - \sin x] dx \\
&= \tfrac{1}{2}\left[-\tfrac{1}{9}\cos 9x + \cos x\right] + c \\
&= \tfrac{1}{2}\cos x - \tfrac{1}{18}\cos 9x + c
\end{aligned}$$

Exercise Set 3.2

Evaluate the integrals.

1. $\int_0^{\pi/2} \cos^2\theta \, d\theta$

2. $\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx$

3. $\int \tan^2 x \, dx$

4. $\int \frac{\sin\phi}{\cos^3\phi} d\phi$

5. $\int_{\pi/6}^{\pi/2} \cot^2 x \, dx$

6. $\int \operatorname{cosec} x \, dx$

7. $\int \sin 2\theta \sin 6\theta \, d\theta$

8. $\int_0^{\pi/2} \cos 5t \cos 10t \, dt$

9. $\int_0^{\pi/6} \sqrt{1+\cos 2x} \, dx$

10. $\int_0^{\pi/4} \sqrt{1-\cos 4\theta} \, d\theta$

11. $\int \cos^3 at \, dt$

12. $\int \sin^3 x \cos^3 x \, dx$

13. $\int \sin^2 x \cos^4 x \, dx$

14. $\int \sin x \cos\left(\frac{x}{2}\right) dx$

15. $\int \sin^2 3x \, dx$

16. $\int \frac{1}{\sqrt{1+\cos 2x}} dx$

[1-10] [Ref: Calculus – J. Stewart – 8th ed.: Page: 484], **[11-16]** [Ref: Calculus – H. Anton – 10th ed.: Page: 507]

Integration by Trigonometric Substitutions: In finding the area of a circle or an ellipse, an integral of the form $\int \sqrt{a^2 - x^2}dx$ arises, where $a > 0$. We require techniques with integrals that contain expressions of the form, $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$ and $\sqrt{x^2 + a^2}$.

Expression in the integrand	Substitution	Simplification
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta} = \sqrt{a^2(1 - \sin^2\theta)} = \sqrt{a^2\cos^2\theta} = a\lvert\cos\theta\rvert$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$\sqrt{a^2 + x^2} = \sqrt{a^2 + a^2\tan^2\theta} = \sqrt{a^2(1 + \tan^2\theta)} = \sqrt{a^2\sec^2\theta} = a\lvert\sec\theta\rvert$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2} = \sqrt{a^2(\sec^2\theta - 1)} = \sqrt{a^2\tan^2\theta} = a\lvert\tan\theta\rvert$

Example 8: Evaluate $\int \frac{\sqrt{9-x^2}}{x^2}dx$.

Solution: Let, $x = 3\sin\theta$, So, $dx = 3\cos\theta d\theta$. Again, $x^2 = 9\sin^2\theta$. We also have, $\sin\theta = \frac{x}{3}$

Thus, $\sqrt{9 - x^2} = \sqrt{9 - 9\sin^2\theta} = \sqrt{9\cos^2\theta} = 3|\cos\theta| = 3\cos\theta$ ($\cos\theta \geq 0$, as $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$)

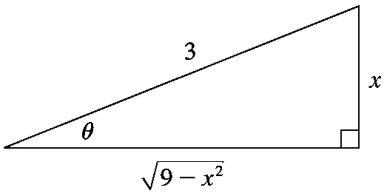
$\int \frac{\sqrt{9-x^2}}{x^2}dx = \int \frac{3\cos\theta}{9\sin^2\theta}3\cos\theta d\theta$

$= \int \frac{\cos^2\theta}{\sin^2\theta}d\theta = \int \cot^2\theta\, d\theta$

$= \int(\operatorname{cosec}^2\theta - 1)d\theta$

$= -\cot\theta - \theta + c$

Since $\sin\theta = \frac{x}{3}$, we label the opposite side and the hypotenuse of the triangle as having lengths x and 3. Then the Pythagorean Theorem gives the length of the adjacent side as $\sqrt{9 - x^2}$.



So, $\cot\theta = \frac{\sqrt{9-x^2}}{x}$. Since $\sin\theta = \frac{x}{3}$, we have $\theta = \sin^{-1}\left(\frac{x}{3}\right)$.

Finally, $\int \frac{\sqrt{9-x^2}}{x^2}dx = -\frac{\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + c$

Example 9: Find $\int \frac{1}{x^2\sqrt{x^2+4}}dx$.

Solution: At Denominator, we have $x^2\sqrt{x^2 + 4}$. Now, $\sqrt{x^2 + 4}$ can be written as $\sqrt{x^2 + 2^2}$.

Let, $x = 2\tan\theta$ (where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$), So, $dx = 2\sec^2\theta d\theta$. Again, $x^2 = 4\tan^2\theta$. We also have, $\tan\theta = \frac{x}{2}$.

Thus, $\sqrt{x^2 + 4} = \sqrt{4\tan^2\theta + 4} = \sqrt{4\sec^2\theta} = 2|\sec\theta| = 2\sec\theta$ ($\sec\theta \geq 0$, as $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$)

$\int \frac{1}{x^2\sqrt{x^2+4}}dx = \int \frac{1}{4\tan^2\theta \cdot 2\sec\theta}2\sec^2\theta d\theta$

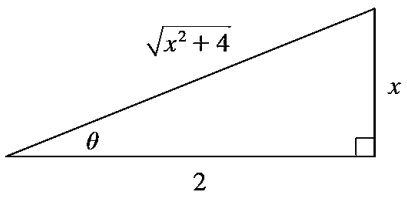
$= \frac{1}{4}\int \frac{\sec\theta}{\tan^2\theta}d\theta = \frac{1}{4}\int \frac{1}{\cos\theta}\cdot\frac{\cos^2\theta}{\sin^2\theta}d\theta$

$= \frac{1}{4}\int \frac{\cos\theta}{\sin^2\theta}d\theta = \frac{1}{4}\int \frac{dz}{z^2}$

$= \frac{1}{4}\cdot\left(-\frac{1}{z}\right) + c = -\frac{1}{4\sin\theta} + c = -\frac{\operatorname{cosec}\theta}{4} + c$

Let, $\sin\theta = z$. So, $\cos\theta\, d\theta = dz$

Since $\tan\theta = \frac{x}{2}$, we label the opposite side and the adjacent side of the triangle as having lengths x and 2. Then the Pythagorean Theorem gives the length of the hypotenuse as $\sqrt{x^2 + 4}$. So, $\operatorname{cosec}\theta = \frac{\sqrt{x^2+4}}{x}$



Finally, $\int \frac{1}{x^2\sqrt{x^2+4}}dx = -\frac{\sqrt{x^2+4}}{4x} + c$

Exercise 3.3

Evaluate the integrals.

1. $\int \frac{dx}{x^2\sqrt{4-x^2}}dx$

2. $\int \frac{x^3}{\sqrt{x^2+4}}dy$

3. $\int \frac{\sqrt{x^2-4}}{x}dx$

4. $\int \frac{x^2}{\sqrt{9-x^2}}dx$

5. $\int_0^3 \frac{x}{\sqrt{36-x^2}}dx$

6. $\int_0^{1/2} x\sqrt{1-4x^2}dx$

7. $\int_0^2 \frac{dt}{\sqrt{4+t^2}}$

8. $\int \frac{\sqrt{x^2-9}}{x^3}dx$

9. $\int_0^1 \frac{dx}{(x^2+1)^2}$

10. $\int_0^{0.6} \frac{x^2}{\sqrt{9-25x^2}}dx$

11. $\int \frac{dx}{\sqrt{x^2+2x+5}}$

12. $\int \sqrt{9-x^2}\,dx$

13. $\int \frac{x^2}{\sqrt{7+x^2}}dx$

14. $\int \frac{3x^3}{\sqrt{1-x^2}}dx$

15. $\int \frac{\sqrt{16-x^2}}{x}dx$

16. $\int \sqrt{1-9x^2}dx$

17. $\int x\sqrt{1-x^2}dx$

[1-11] [Ref: Calculus – J. Stewart – 8th Ed. Page: 491], **[12-17]** [Ref: Calculus – H Anton – 10th Ed. Page: 513]

Integration by Partial Fractions: The method of breaking down rational functions into sum of simpler rational functions is known as the Method of Partial Fraction.

Example 10: Evaluate $\int \frac{x^2+2x-1}{2x^3+3x^2-2x}\,dx$

Solution: We factor the denominator as

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2) = x(2x - 1)(x + 2)$$

The partial fraction decomposition of the integrand has the form,

$$\frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

$$x^2 + 2x - 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

$$x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

Equating coefficients gives the following system of equations for A, B, and C:

$$\begin{aligned} 2A + B + 2C &= 1 \\ 3A + 2B - C &= 2 \\ -2A &= -1 \end{aligned}$$

Solving we get, $A = \frac{1}{2}$, $B = \frac{1}{5}$, $C = -\frac{1}{10}$. Now,

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x}\,dx = \int \left[\frac{1}{2}\cdot\frac{1}{x} + \frac{1}{5}\cdot\frac{1}{2x - 1} + \left(-\frac{1}{10}\right)\cdot\frac{1}{x + 2}\right] dx$$

$$= \int \left[\frac{1}{2}\frac{1}{x} + \frac{1}{5}\cdot\frac{1}{2x - 1} - \frac{1}{10}\cdot\frac{1}{x + 2}\right] dx$$

$$= \frac{1}{2}\ln|x| + \frac{1}{5}\cdot\frac{1}{2}\ln|2x - 1| - \frac{1}{10}\ln|x + 2| + c$$

$$= \frac{1}{2}\ln|x| + \frac{1}{10}\ln|2x - 1| - \frac{1}{10}\ln|x + 2| + c$$

Exercise Set 3.4

Evaluate the integrals.

1. $\int \frac{5x+1}{(2x+1)(x-1)}\,dx$

2. $\int \frac{y}{(y+4)(2y-1)}\,dy$

3. $\int_0^1 \frac{2}{2x^2+3x+1}\,dx$

4. $\int_0^1 \frac{x-4}{x^2-5x+6}\,dx$

5. $\int \frac{1}{(x+a)(x+b)}\,dx$

6. $\int_1^2 \frac{4y^2-7y-12}{y(y+2)(y-3)}\,dy$

7. $\int_0^1 \frac{x^2+x+1}{(x+1)^2(x+2)}\,dx$

8. $\int_2^3 \frac{x(3-5x)}{(3x-1)(x-1)^2}\,dx$

9. $\int \frac{10}{(x-1)(x^2+9)}\,dx$

10. $\int \frac{x^2-x+6}{x^3+3x}\,dx$

11. $\int \frac{dx}{x^2-3x-10}$

12. $\int \frac{2x^2+3}{x(x-1)^2}\,dx$

13. $\int \frac{2x^2-2x-1}{x^3-x^2}\,dx$

14. $\int \frac{2x^2-1}{(4x-1)(x^2+1)}\,dx$

15. $\int \frac{4x+17}{x^2+6x-16}\,dx$

16. $\int \frac{dx}{(x+2)(x-1)^2}$

17. $\int_4^5 \frac{x-1}{(x-3)(x-2)}\,dx$

[1-10] [Ref: Calculus – J. Stewart – 8th Ed. Page: 501], **[11-17]** [Ref: Calculus – H. Anton – 10th Ed. Page: 521]

Chapter 4

Application of Integration: Area

Area Between the curves:

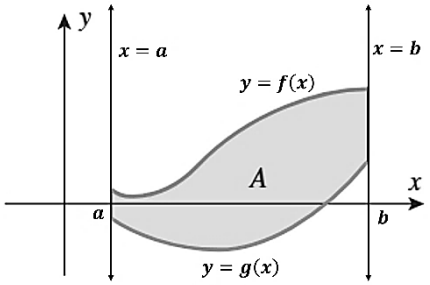


Figure T1.1

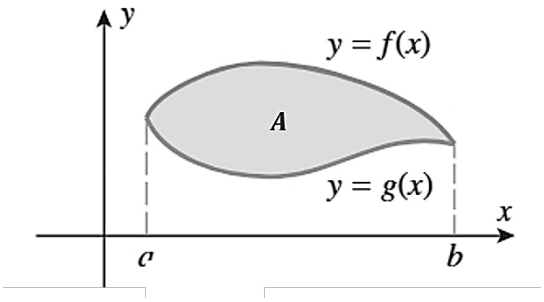


Figure T1.2

Suppose that f and g are continuous functions on an interval $[a, b]$ and $f(x) \geq g(x)$ for $a \leq x \leq b$. [This means that the curve $y = f(x)$ lies **above** the curve $y = g(x)$].

So, The area $\boldsymbol{A = \int_a^b [f(x) - g(x)]dx}$

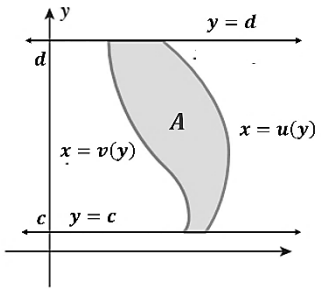


Figure T2.1

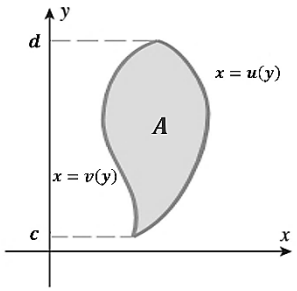


Figure T2.2

Suppose that u and v are continuous functions on an interval $[c, d]$ and $u(x) \geq v(x)$ for $c \leq y \leq d$. [This means that the curve $x = u(y)$ lies **right of** the curve $x = v(y)$].

So, The area $\boldsymbol{A = \int_c^d [u(y) - v(y)]dy}$.

Example 1: Sketch and find the area of the region bounded above by $y = e^x$, bounded below by $y = x$, and bounded on the sides by
$x = 0$ and $x = 1$

Solution: The region is shown in Figure 1. The upper boundary curve is $y = e^x$ and the lower boundary curve is $y = x$. So, we have $f(x) = e^x$, $g(x) = x$, $a = 0$, and $b = 1$.

$$A = \int_0^1 (e^x - x)\,dx = \left[e^x - \frac{x^2}{2}\right]_0^1$$
$$= e^1 - \frac{1}{2} - 1$$
$$= e - 1.5 \approx 1.21828$$

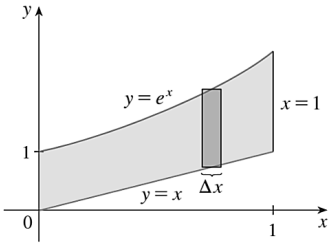


Figure 4.1.1

Example 2: Sketch and find the area of the region bounded above by $y = x + 6$, bounded below by $y = x^2$, and bounded on the sides by the lines $x = 0$ and $x = 2$.

Solution: The region is shown in Figure 2. The upper boundary curve is $y = x + 6$ and the lower boundary curve is $y = x^2$. So, we have $f(x) = x + 6$, $g(x) = x^2$, $a = 0$, and $b = 2$.

$$A = \int_0^2 (x + 6 - x^2)\,dx$$
$$= \left[\frac{x^2}{2} + 6x - \frac{x^3}{3}\right]_0^1$$
$$= \left(\frac{1}{2} + 6 - \frac{1}{3}\right) - (0 + 0 - 0)$$
$$= \frac{34}{3} - 0$$

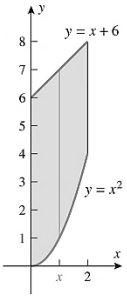


Figure 4.2.1

Example 3: Sketch and find the area of the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$

Solution: The points of intersection of the parabolas by solving $x^2 = 2x - x^2$, so we have $x = 0$ and $x = 1$. Hence the points of intersection are (0,0), and (1,1).

We see from Figure 3 that the top and bottom boundaries are, $y_T = 2x - x^2$ and $y_B = x^2$

$$A = \int_0^1 (y_T - y_B)\,dx$$
$$= \int_0^1 (2x - x^2 - x^2)\,dx = \int_0^1 (2x - 2x^2)\,dx = 2\int_0^1 (x - x^2)\,dx$$
$$= 2\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 2\left(\frac{1}{2} - \frac{1}{3}\right) = 2 \cdot \frac{1}{6} = \frac{1}{3}$$

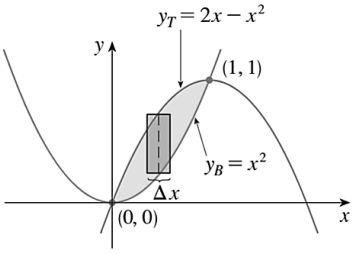


Figure 4.3.1

Example 4: Sketch and find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

Solution: We can find intersections by equating the expressions for y. By solving we get the points of intersection are $(-1, -2)$ and $(5, 4)$. So, we integrate between the appropriate y-values, $u(x) = y + 1$, $v(x) = \frac{1}{2}y^2 - 3$, $a = -2$, $b = 4$

$$A = \int_{-2}^4 (x_R - x_L)\,dy$$
$$= \int_{-2}^4 \left[(y + 1) - \left(\frac{1}{2}y^2 - 3\right)\right] dy = \int_{-2}^4 \left(-\frac{1}{2}y^2 + y + 4\right) dy$$
$$= \left[-\frac{1}{2}\frac{y^3}{3} + \frac{y^2}{2} + 4y\right]_{-2}^4$$
$$= \left[-\frac{1}{6}(64) + 8 + 16\right] - \frac{1}{6}(-8) + 2 - 8 = 18$$

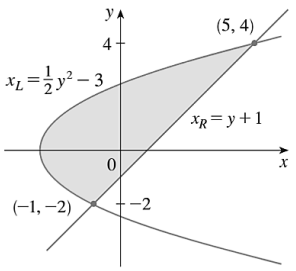


Figure 4.4.1

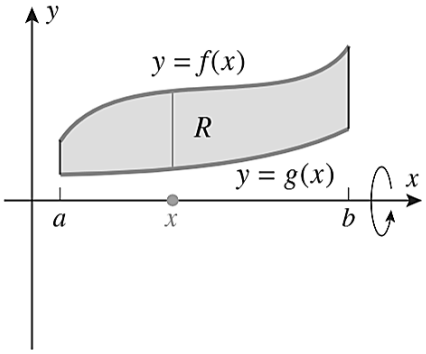
Exercise Set 4.1

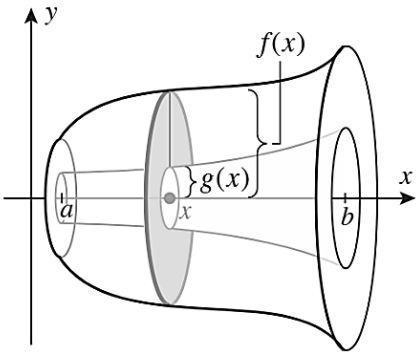
Sketch the region enclosed by the given curves and find its area.

1. $y = e^x, y = x^2 - 1, x = -1, x = 1.$
2. $y = \sin x, y = x, x = \frac{\pi}{2}, x = \pi.$
3. $y = (x-2)^2, y = x.$
4. $y = x^2 - 4x, y = 2x.$
5. $y = 12 - x^2, y = x^2 - 6.$
6. $y = x^2, y = 4x - x^2.$
7. $x = 2y^2, x = 4 + y^2.$
8. $y = \sqrt{x-1}, x - y = 1.$
9. $y = x^3, y = x.$
10. $y = x^2, y = \sqrt{x}. x = \frac{1}{4}, x = 1.$
11. $y = \cos 2x, y = 0, x = -\frac{\pi}{4}, x = \frac{\pi}{4}.$
12. $x^2 = y, x = y - 2.$

[1-9] [Ref: Calculus – J. Stewart – 8th Ed.: Page: 434], **[10-12]** [Ref: Calculus – H. Anton – 10th Ed.: Page: 419]

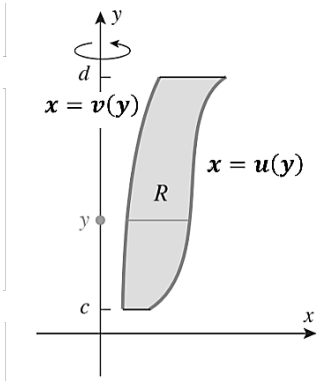
Application of Integration: Volume

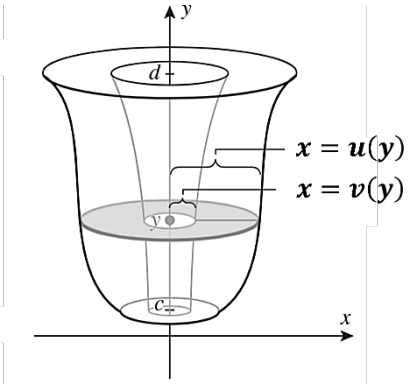




Let f and g be continuous and nonnegative on $[a, b]$, and suppose that $f(x) \geq g(x)$ for all x in the interval $[a, b]$. Let R be the region that is bounded above by $y = f(x)$, below by $y = g(x)$, and on the sides by the lines $x = a$ and $x = b$. Thus, the volume of the solid is,

$$\boldsymbol{V = \int_a^b \pi\left([f(x)]^2 - [g(x)]^2\right)dx}$$





Let u and v be continuous and nonnegative on $[c, d]$, and suppose that u$(y) \geq v(y)$ for all y in the interval $[c, d]$. Let R be the region that is bounded on left by $x = v(y)$, right by $x = u(y)$, and on the sides by the lines $y = c$ and $y = d$. Thus, the volume of the solid is,

$$\boldsymbol{V = \int_c^d \pi\left([u(y)]^2 - [v(y)]^2\right)dy}$$

Example 5: Find the volume of the solid obtained by rotating about the x-axis the region under the curve $y = \sqrt{x}$ from 0 to 1.
Solution:

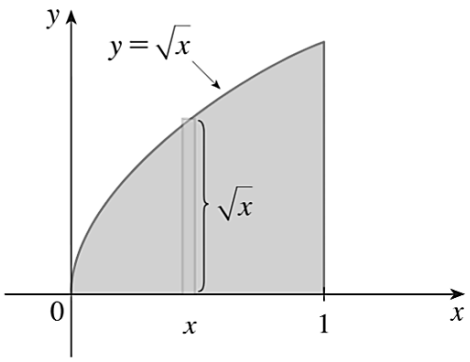


Figure 4.5.1

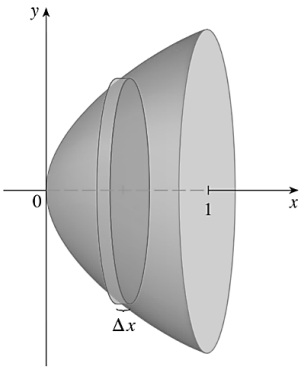


Figure 4.5.2

The region is shown in Figure 1.1. If we rotate about the x-axis, we get the solid shown in Figure 1.2. The area of the cross section of the solid will be $A(x) = \pi\left(\sqrt{x}\right)^2 = \pi x$. Given, the solid lies in $x = 0$ and $x = 1$. So, the volume will be,

$V = \int_0^1 A(x)\,dx$

$= \int_0^1 \pi x\,dx$

$= \pi\left[\frac{x^2}{2}\right]_0^1$

$= \pi\left[\frac{1}{2} - 0\right] = \mathbf{\frac{\pi}{2}}$

Example 6: Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$, and $x = 0$ about the y-axis.
Solution:

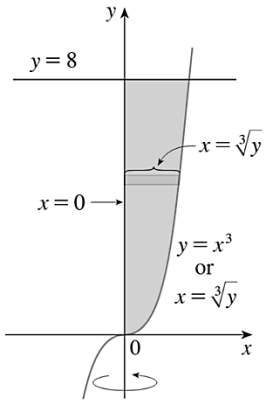


Figure 4.6.1

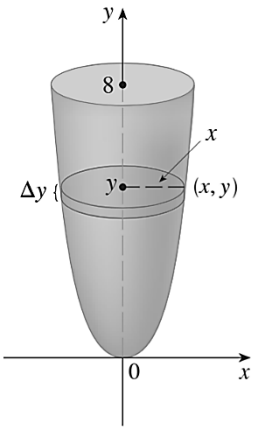


Figure 4.6.2

The region is shown in Figure 4.6.1, and the region is rotated about the y-axis, which gives the solid is shown in Figure 4.6.2. So, the area of a cross-section through y is $A(y) = \pi\left(\sqrt[3]{y}\right)^2 = \pi y^{\frac{2}{3}}$

Given, the solid lies between $y = 0$ and $y = 8$. So, the volume will be,

$V = \int_0^8 A(y)\,dy$

$= \int_0^8 \pi y^{\frac{2}{3}}\,dy$

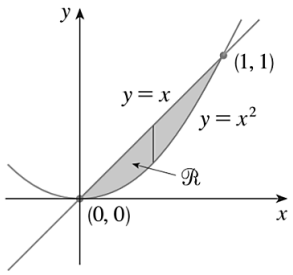
$= \pi\left[\frac{3}{5} \cdot y^{\frac{5}{3}}\right]_0^8$

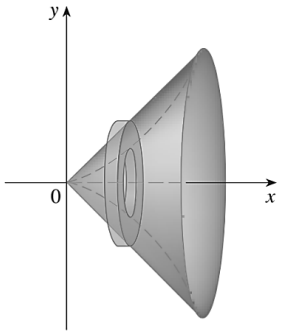
$= \frac{3\pi}{5}\left[8^{\frac{5}{3}} - 0\right]$

$= \frac{3\pi}{5} \cdot 32 = \mathbf{\frac{96\pi}{5}}$.

Example 7: The region $\mathfrak{R}$ is enclosed by the curves $y = x$ and $y = x^2$ is rotated about the x-axis. Find the volume of the resulting solid.

Solution:





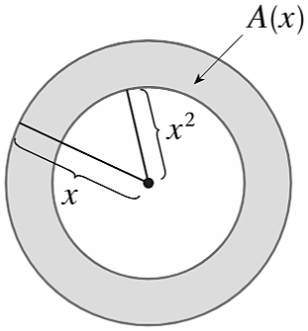


Figure 4.7.1 Figure 4.7.2 Figure 4.7.3

The curves $y = x$ and $y = x^2$ intersect at the points (0,0) and (1,1). The region between them, the solid of rotation, and a cross-section perpendicular to the x-axis are shown in Figure 3.1, Figure 3.2 and Figure 3.3. We find the cross-sectional area by subtracting the area of the inner circle from the area of the outer circle:

$A(x) = \pi(x)^2 - \pi(x^2)^2 = \pi x^2 - \pi x^4 = \pi(x^2 - x^4)$

From the intersecting points, the solid lies between $x = 0$ and $x = 1$. So, the volume will be,

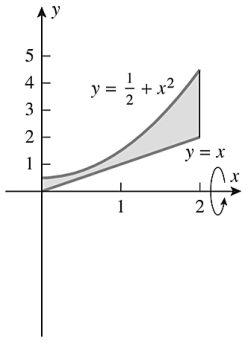
$V = \int_0^1 A(x)\,dx$

$= \int_0^1 \pi(x^2 - x^4)\,dx = \pi\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1$

$= \pi\left[\frac{1}{3} - \frac{1}{5} - (0 - 0)\right] = \mathbf{\frac{2\pi}{15}}$.

Example 8: Find the volume of the solid generated when the region between the graphs of the equations $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ over the interval $[0, 2]$ is revolved about the x-axis.

Solution:



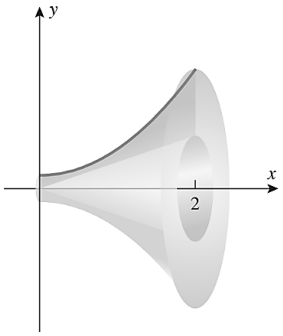


Figure 4.8.1 Figure 4.8.2

Given curves, $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$. The region between them, the solid of rotation, are shown in Figure 4.1 and Figure 4.2. We find the cross-sectional area:

$A(x) = \pi\left(\frac{1}{2} + x^2\right)^2 - \pi(x)^2 = \frac{\pi}{4} + \pi x^2 + \pi x^4 - \pi x^2 = \pi x^4 + \frac{\pi}{4}$

Given, the solid lies between $x = 0$ and x= 2. So, the volume will be,

$V = \int_0^2 A(x)\,dx$

$= \int_0^2 \left(\pi x^4 + \frac{\pi}{4}\right) dx = \pi\left[\frac{x^5}{5} + \frac{x}{4}\right]_0^2$

$= \pi\left[\frac{32}{5} + \frac{1}{2} - (0 + 0)\right] = \mathbf{\frac{69\pi}{10}}$.

Exercise Set 4.2

Sketch the region and the solid. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

1. $y = x + 1, y = 0, x = 0, x = 2$; about x-axis.

2. $y = \frac{1}{x}, y = 0, x = 1, x = 4$; about x-axis.

3. $y = \sqrt{x - 1}, y = 0, x = 5$; about x-axis.

4. $y = e^x, y = 0, x = -1, x = 1$; about x-axis.

5. $x = 2\sqrt{y}, x = 0, x = 9$; about y-axis.

6. $2x = y^2, x = 0, y = 4$; about y-axis.

7. $y = x^3, y = x, x \geq 0$; about x-axis.

8. $y = 6 - x^2, y = 2$; about x-axis.

9. $y^2 = x, x = 2y$; about y-axis.

10. $x = 2 - y^2, x = y^4$; about y-axis.

11. $y = x^2, x = y^2$; about y-axis.

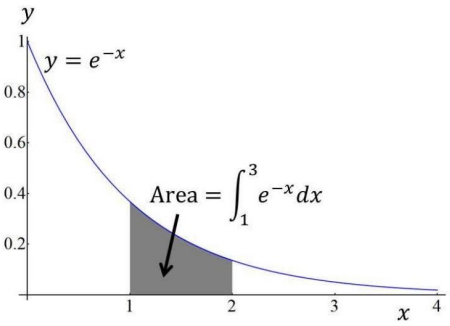
12. $x = 1 - y^2, x = 2 + y^2, y = -1, y = 1$; about y-axis.

[1-9] [Ref: Calculus – J. Stewart – 8th Ed.: Page: 446]**, [11-12]** [Ref: Calculus – H Anton – 10th Ed.: Page: 429]

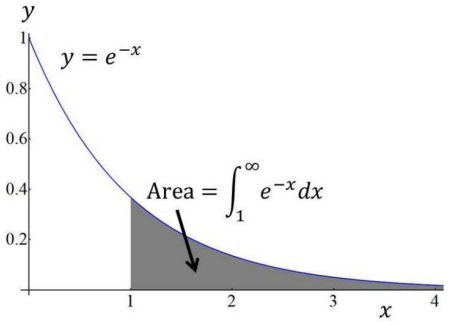
Chapter 5

Improper Integrals, Gamma and Beta Functions

Improper Integrals: An improper integral is an extended concept of a definite integral that has infinite limits on one or both ends of the interval and/or an integrand that becomes infinite at one or more points within the interval of integration (Fig 1). Improper integral is called convergent if the limit of the integral exists with finite value and divergent if the limit of the integral does not exist or has infinite value.



Definite Integral



Improper Integral

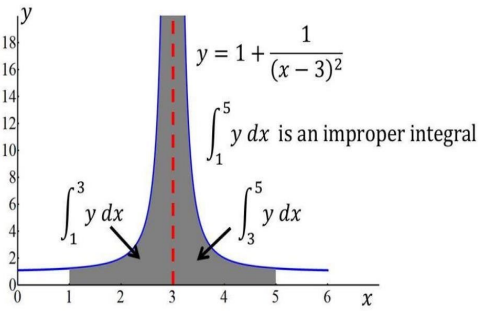


Figure 5.1: The above figures show the geometrical concept of the definite integral and improper integral.

Improper integral with infinite limit

(1) $\int_a^{\infty} f(x)dx = \lim_{b\to\infty} \int_a^b f(x)dx$

(2) $\int_{-\infty}^{b} f(x)dx = \lim_{a\to-\infty} \int_a^b f(x)dx$

(3) $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{c} f(x)dx + \int_c^{\infty} f(x)dx = \lim_{a\to-\infty} \int_a^c f(x)dx + \lim_{b\to\infty} \int_c^b f(x)dx$; where c is any convenient point.

Example 1: $\int_1^{\infty} \frac{dx}{x^3}$

Solution: $\int_1^{\infty} \frac{dx}{x^3}$

$$= \lim_{b\to\infty} \int_1^b \frac{dx}{x^3} = \lim_{b\to\infty} \left[-\frac{1}{2x^2}\right]_1^b = \lim_{b\to\infty} \left[-\frac{1}{2b^2} + \frac{1}{2}\right] = \mathbf{\frac{1}{2}}$$

Example 2: $\int_{-\infty}^{0} \frac{xdx}{1+x^2}$

Solution: $\int_{-\infty}^{0} \frac{xdx}{1+x^2}$

$$= \lim_{a\to-\infty} \int_a^0 \frac{xdx}{1+x^2} = \lim_{a\to-\infty} \left[\frac{1}{2}\ln(x^2+1)\right]_a^0 = \lim_{a\to-\infty} \left[-\frac{1}{2}\ln(a^2+1)\right] \to -\infty$$

Hence the integral diverges.

Example 3: $\int_{-\infty}^{\infty} \frac{dx}{1+(x-1)^2}$

Solution: $\int_{-\infty}^{\infty} \frac{dx}{1+(x-1)^2}$

$$= \int_{-\infty}^{1} \frac{dx}{1+(x-1)^2} + \int_1^{\infty} \frac{dx}{1+(x-1)^2} = \lim_{a\to-\infty} \int_a^1 \frac{dx}{1+(x-1)^2} + \lim_{b\to\infty} \int_1^b \frac{dx}{1+(x-1)^2}$$

$$= \lim_{a\to-\infty} [\tan^{-1}(x-1)]_a^1 + \lim_{b\to\infty} [\tan^{-1}(x-1)]_1^b = \lim_{a\to-\infty} [0 - \tan^{-1}(a-1)] + \lim_{b\to\infty} [\tan^{-1}(b-1) - 0]$$

$$= -\left[-\frac{\pi}{2}\right] + \left[\frac{\pi}{2}\right] = \boldsymbol{\pi}$$

Improper integral with infinite integrand

(1) Infinite at $x = a$: $\int_a^b f(x)dx = \lim_{t\to a^+} \int_t^b f(x)dx$

(2) Infinite at $x = b$: $\int_a^b f(x)dx = \lim_{t\to b^-} \int_a^t f(x)dx$

(3) Infinite at $x = c$ where $a < c < b$: $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

$= \lim_{t\to c^-} \int_a^t f(x)dx + \lim_{t\to c^+} \int_t^b f(x)dx$

Example 4: $\int_0^1 \frac{dx}{\sqrt{x}}$

Solution: $\int_0^1 \frac{dx}{\sqrt{x}}$

$= \lim_{a\to 0^+} \int_a^1 x^{-1/2}dx$

$= \lim_{a\to 0^+} \left[2x^{1/2}\right]_a^1$

$= \lim_{a\to 0^+} [2 - 2\sqrt{a}] = \mathbf{2}$

Example 5: $\int_0^2 \frac{dx}{\sqrt{4-x^2}}$

Solution: $\int_0^2 \frac{dx}{\sqrt{4-x^2}}$

$= \lim_{b\to 2^-} \int_0^b \frac{dx}{\sqrt{4-x^2}}$

$= \lim_{b\to 2^-} \left[\sin^{-1}\left(\frac{x}{2}\right)\right]_0^b$

$= \lim_{b\to 2^-} \sin^{-1}\left(\frac{b}{2}\right)$

$= \frac{\boldsymbol{\pi}}{\mathbf{2}}$

Alternatively,

Let, $x = 2\sin\theta$

$\Rightarrow dx = 2\cos\theta d\theta$

x	0	2^-
θ	0	$\frac{\pi^-}{2}$

Thus, $\int_0^2 \frac{dx}{\sqrt{4-x^2}}$

$= \int_0^{\pi/2} \frac{2\cos\theta d\theta}{2\cos\theta}$

$= \int_0^{\pi/2} d\theta = [\theta]_0^{\pi/2} = \frac{\boldsymbol{\pi}}{\mathbf{2}}$

Exercise Set 5.1

1. Find, when they exist, the value of the following integrals:

(a) $\int_1^\infty \frac{1}{x^2}dx$

(b) $\int_{-\infty}^0 \frac{1}{\sqrt{3-x}}dx$

(c) $\int_{-\infty}^\infty xe^{-x^2}dx$

(d) $\int_0^3 \frac{1}{\sqrt{3-x}}dx$

(e) $\int_0^\infty \frac{1}{1+x^2}dx$

(f) $\int_0^1 \frac{1}{x^2}dx$

Answers: (a) 1 (b) Divergent (c) 0 (d) $2\sqrt{3}$ (e) $\frac{\pi}{2}$ (f) Divergent

The Gamma function: The gamma function $\Gamma(n)$ is defined by $\boldsymbol{\Gamma(n+1) = \int_0^\infty x^n e^{-x} dx}$

where, for convergence of the integral, $n > 0$.

Integration by parts gives,

$$\Gamma(n+1) = \int_0^\infty x^n e^{-x} dx$$
$$= [-x^n e^{-x}]_0^\infty + n\int_0^\infty x^{n-1} e^{-x} dx$$
$$= n\Gamma(n)$$

So, $\boldsymbol{\Gamma(n+1) = n\Gamma(n)}$; when $n > 0$

By using this recurrence relation when n is an integer and $n \geq 1$, we have

$$\Gamma(n+1) = n\Gamma(n)$$
$$= n(n-1)\Gamma(n-1) = \cdots = n!\,\Gamma(1)$$

But $\Gamma(1) = \int_0^\infty e^{-x} dx = 1$.

So, $\boldsymbol{\Gamma(n+1) = n!}$; when n is an integer and $n \geq 1$.

From definition,

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2} e^{-1} dt$$
$$= 2\int_0^\infty e^{-u^2} du \qquad \text{By substituting } t = u^2$$
$$= \sqrt{\pi} \qquad \textit{[For evaluation of the integral advanced idea is needed and quote the result without proof.]}$$

So, $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$

Example 6: $\Gamma\left(\frac{5}{2}\right)$

Solution: $\Gamma\left(\frac{5}{2}\right) = \Gamma\left(\frac{3}{2}+1\right)$
$$= \frac{3}{2}\Gamma\left(\frac{3}{2}\right) = \frac{3}{2}\Gamma\left(\frac{1}{2}+1\right)$$
$$= \frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \boldsymbol{\frac{3}{4}\sqrt{\pi}}$$

Example 7: Evaluate the integral $\int_0^\infty \frac{e^{-4x}}{x^{1/2}} dx$.

Solution: $\int_0^\infty \frac{e^{-4x}}{x^{1/2}} dx$
$$= \int_0^\infty \frac{2}{u^{1/2}} e^{-u} \frac{du}{4}$$
$$= \frac{1}{2}\int_0^\infty u^{-1/2} e^{-u} du$$
$$= \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \boldsymbol{\frac{1}{2}\sqrt{\pi}}$$

Let, $4x = u$. So, $x = \frac{u}{4}$
$\Rightarrow\ dx = \frac{du}{4}$

x	0	∞
u	0	∞

Gamma Function for negative values

The recurrence relation for positive values of x is extended in defining the gamma function for negative values of x. For $x < 0$, the recurrence relation is written as

$$\boldsymbol{\Gamma(x) = \frac{\Gamma(x+1)}{x}}$$

From the above definition it becomes infinity at $x = 0$, and hence $\Gamma(x)$ is not defined at all negative values of x. It is important to note that $\Gamma(x)$ for negative values is not defined by the integral.

Example 8: $\Gamma\left(-\frac{5}{2}\right)$

Solution: $\Gamma\left(-\frac{5}{2}\right) = \frac{\Gamma\left(-\frac{3}{2}\right)}{\left(-\frac{5}{2}\right)} = \frac{\Gamma\left(-\frac{1}{2}\right)}{\left(-\frac{5}{2}\right)\left(-\frac{3}{2}\right)} = \frac{\Gamma\left(\frac{1}{2}\right)}{\left(-\frac{5}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{1}{2}\right)}$
$$= \boldsymbol{\frac{8}{15}\sqrt{\pi}}$$

The Beta function: The beta-function $\mathrm{B}(m,n)$ is defined by the integral

$$\mathbf{B(m+1,n+1)} = \int_0^1 \boldsymbol{x^m(1-x)^n dx}; \qquad (m>0, n>0)$$

$$\begin{aligned}\mathrm{B}(n+1,m+1) &= \int_0^1 x^n(1-x)^m dx \\ &= \int_1^0 (1-u)^n u^m(-du) \\ &= \int_0^1 (1-u)^n\, u^m\, du \\ &= \mathrm{B}(m+1,n+1)\end{aligned}$$

Let, $1-x=u$. So, $x=1-u$
$\Rightarrow\ dx=-du$

x	0	1
u	1	0

That is, $\mathrm{B}(m,n)=\mathrm{B}(n,m)$. Thus beta-function is symmetric in m,n.
An alternative form of the beta-function, obtained from the definition by putting $x=\sin^2\theta$, is

$$\mathrm{B}(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\,\theta\cos^{2n-1}\,\theta d\theta$$

Relation between Gamma and Beta-Functions

The relation between Gamma- and Beta-Functions may be written as

$$\mathbf{B(m,n)} = \frac{\boldsymbol{\Gamma(m)\Gamma(n)}}{\boldsymbol{\Gamma(m+n)}} \qquad \cdots(\mathbf{2})$$

The proof of the relation need advanced ideas and is not included here.
Gamma- and Beta-Functions may be used in evaluating integrals. For example, from eq. (2), we get

$$\int_0^{\pi/2} \sin^p x\cos^q x dx = \frac{1}{2}\,\mathrm{B}\left(\frac{p+1}{2},\frac{q+1}{2}\right) == \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$$

Example 9: Evaluate $B(3,5)$

Solution: $B(3,5) = \frac{\Gamma(3)\Gamma(5)}{\Gamma(3+5)}$
$= \frac{(2!)(4!)}{(7!)}$
$= \mathbf{\frac{1}{105}}$

Example 10: Evaluate $\mathrm{B}\left(\frac{7}{2},4\right)$

Solution: $\mathrm{B}\left(\frac{7}{2},4\right) = \frac{\Gamma(\frac{7}{2})\Gamma(4)}{\Gamma(\frac{7}{2}+4)}$
$= \frac{\frac{5}{2}.\frac{3}{2}.\frac{1}{2}.\sqrt{\pi}\,.\,3\,.\,2\,.\,1}{\frac{13}{2}.\frac{11}{2}.\frac{9}{2}.\frac{7}{2}.\frac{5}{2}.\frac{3}{2}.\frac{1}{2}.\sqrt{\pi}} = \mathbf{\frac{32}{3003}}$

Example 11: Evaluate the integral $\int_0^2 \frac{x^2}{\sqrt{2x-x^2}}dx$.

Solution: $\int_0^2 \frac{x^2}{\sqrt{2x-x^2}}dx = \int_0^2 \frac{x^2}{\sqrt{2x(1-\frac{x}{2})}}dx = \frac{1}{\sqrt{2}}\int_0^2 x^{3/2}\left(1-\frac{x}{2}\right)^{-1/2}dx$
$= \frac{1}{\sqrt{2}}\int_0^1 (2u)^{\frac{3}{2}}(1-u)^{-\frac{1}{2}}2du = 4\int_0^1 u^{\left(\frac{5}{2}-1\right)}(1-u)^{\frac{1}{2}-1}du$
$= 4\,\mathrm{B}\left(\frac{5}{2},\frac{1}{2}\right) = 4\frac{\Gamma\left(\frac{5}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(3)} = 4\frac{\left(\frac{3}{2}\right)\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{2} = \frac{3\pi}{2}$

Let, $\frac{x}{2}=u$. So, $x=2u$
$\Rightarrow\ dx=2du$

x	0	2
u	0	1

Example 12: Evaluate the integral $\int_0^{\pi/2} \sin^{5/2}x\cos^3 x dx$.

Solution: Using Beta function, $\int_0^{\pi/2} \sin^{5/2}x\cos^3 x dx = \frac{1}{2}\mathrm{B}\left(\frac{\frac{5}{2}+1}{2},\frac{3+1}{2}\right) = \frac{1}{2}\mathrm{B}\left(\frac{\frac{7}{2}}{2},\frac{4}{2}\right) = \frac{1}{2}\mathrm{B}\left(\frac{7}{4},2\right)$

$\frac{1}{2}\mathrm{B}\left(\frac{7}{4},2\right) = \frac{\Gamma\left(\frac{7}{4}\right)\Gamma(2)}{2\Gamma\left(\frac{15}{4}\right)} = \frac{\Gamma\left(\frac{7}{4}\right)\times 1}{2\left(\frac{14}{4}\right)\left(\frac{7}{4}\right)\Gamma\left(\frac{7}{4}\right)} = \mathbf{\frac{8}{77}}$

Exercise Set 5.2

1. Evaluate each of the following:

(a) $\Gamma\left(-\frac{3}{2}\right)$

(b) $\frac{\Gamma\left(\frac{5}{2}\right)}{\Gamma\left(\frac{3}{2}\right)}$

(c) $\frac{6\Gamma\left(\frac{8}{3}\right)}{5\Gamma\left(\frac{2}{3}\right)}$

(d) $\mathrm{B}\left(\frac{3}{2}, 2\right)$

2. Evaluate the following:

(a) $\int_0^\infty x^4 e^{-x} dx$

(b) $\int_0^\infty \sqrt{x}\, e^{-x^3} dx$

(c) $\int_0^\infty x^{\frac{3}{2}} e^{-x} dx$

(d) $\int_0^\infty x^6 e^{-2x} dx$

(e) $\int_0^\infty \sqrt{y}\, e^{-y^2} dy$

(f) $\int_0^1 x^4 (1-x)^3 dx$

(g) $\int_0^2 \frac{x^2}{\sqrt{2-x}} dx$

(h) $\int_0^{\pi/2} \cos^6\theta d\theta$

(i) $\int_0^{\pi/2} \sin^8\theta d\theta$

(j) $\int_0^1 y^4\sqrt{1-y^2} dy$

(k) $\int_0^{\pi/2} \sin^7\theta \cos^5\theta d\theta$

(l) $\int_{-\pi/2}^{\pi/2} \sin^6\varphi \cos^4\varphi d\varphi$

Answers

1 (a) $\frac{4}{3}\sqrt{\pi}$ (b) $\frac{3}{2}$ (c) $\frac{4}{3}$ (d) $\frac{4}{15}$

2 (a) 24 (b) $\frac{\sqrt{\pi}}{3}$ (c) $\frac{3}{4}\sqrt{\pi}$ (d) $\frac{45}{8}$ (e) $\frac{1}{2}\Gamma\left(\frac{3}{4}\right)$ (f) $\frac{1}{280}$ (g) $\frac{64\sqrt{2}}{15}$ (h) $\frac{5\pi}{32}$ (i) $\frac{35\pi}{256}$ (j) $\frac{\pi}{32}$ (k) $\frac{1}{120}$ (l) $\frac{3\pi}{256}$